

Lab 1: R review

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Objects

1. Create a code chunk and set the header parameter to TRUE and print out the top rows of the table with `head()` as above.

```
data <- read.table(file =  
  url("http://www2.stat.duke.edu/~pdh10/FCBS/Exercises/azdiabetes.dat"),  
  header = TRUE)  
  
head(data)
```

2. Generate a sequence of 100 equispaced real numbers from 0 to 1 and store it in a variable called `seq2`.

```
seq2 <- seq(from = 0, to = 1, by = 0.01)  
seq2
```

3. Sort the entries in `seq3` from greatest to least.

```
seq3 <- seq(from = -3, to = 3, by = .5)  
sort(seq3, decreasing = TRUE)
```

Matrices

4. Find the variance of each row of `mat5`

```
# generate large matrix
mat5 <- matrix(seq(1,100,1), nrow = 4, ncol = 25, byrow = T)

# apply(X = object, MARGIN = 1 for rows or 2 for columns,
# FUN = function of choice)
# find variance of each row
apply(X = mat5, MARGIN = 1, FUN = var)
```

Distribution Functions

5. Generate 500 samples from a Beta distribution with shape parameter $[a, b] = [0.5, 0.5]$ and store the samples in a variable called W

```
W <- rbeta(500, shape1 = 0.5, shape2 = 0.5)
```

Plotting

6. Use code from above to make a few plots of your own.

```
hist(W,
     main = "Histogram of W",
     xlab = "W", col = "lightblue")

plot(density(W),
     main = "Density Plot of W",
     xlab = "W")

boxplot(W,
     main = "Boxplot of W",
     ylab = "W")
```